



CAUSAL MAPPING TEAMS GLOBALLY – SINCE 2020

📅 20 Feb 2026

Causal mapping teams globally (last 15 years: 2011-2026)

Inclusion rule used here: records from 2011 onward that are clearly causal-mapping work (or close adjacent approaches used to build causal maps), starting from [zotero-bib](#) and then filtering out obvious noise.

Date of most recent publication	Team members	Location / institution	Papers (title, year, citekey)	What they call the approach	Key methods / fit to categorisation / AI use	Primary evidence source mode	Multi-source handling style	Maturity / status	Tooling stack
2026	Vanessa Hammond; Joseph Lea (plus collaborators)	City of Casey, Melbourne (Australia) context; institution not explicit in record	<i>Where to Start? Participatory Systems Mapping for Place-Based Service Integration in the City of Casey</i> (2026) Hammond et al. (2026)	Participatory Systems Mapping (PSM)	Group co-production of causal loop maps; strong fit with group map-building and complexity-aware analysis ; uses network metrics + Action Scales; no direct AI	Group workshops	Aggregated group map with facilitated consensus	Preprint	Workshop facilitation + network analysis

Date of most recent publication	Team members	Location / institution	Papers (title, year, citekey)	What they call the approach	Key methods / fit to categorisation / AI use	Primary evidence source mode	Multi-source handling style	Maturity / status	Tooling stack
					in method				
2025	Steve Powell; Gabriele Caldas Cabral; Fiona Remnant; James Copesta ke; Heather Britt; Rebekah Avar; co-authors	Causal Map / Causal Pathways ecosystem; some records do not specify formal affiliations	AI-assisted Causal Mapping; A Validation Study (2025) (n.d.) ; Qualitative Causal Mapping in Evaluations (2025) Remnant et al. (2025) ; Strengthening Outcome Harvesting with AI-assisted Causal Mapping (2025) Britt et al. (2025) ; Causal Mapping for Evaluators (2024) Powell	Causal mapping ; AI-assisted causal mapping ; causal QDA; QuIP evidence syntheses	Strong fit with document/interview coding, open elicitation, provenance-explicit multi-source synthesis ; explicit separation of evidence assembly vs evaluative judgement; substantial AI support for low-level coding/extraction	Narrative interviews, documents, mixed	Explicit provenance and source-thread logic; map-as-evidence - repository approach	Peer-reviewed + chapter + guidance + report + preprint	Causal Map software + LLM-assisted coding/extraction

Date of most recent publication	Team members	Location / institution	Papers (title, year, citekey)	What they call the approach	Key methods / fit to categorisation / AI use	Primary evidence source mode	Multi-source handling style	Maturity / status	Tooling stack
			et al. (2024); <i>An M&E Time Machine</i> (2024) Powell et al. (2024); <i>Does Our Theory Match Your Theory?</i> (2023) Powell et al. (2023); <i>Measuring the Women's Economic Empowerment... testing QuIP</i> (2022) Avard et al. (2022); <i>Chapter 1 Overview Guide to Causal Mapping</i> (2022) Powell et al. (2022); <i>Guide to</i>						

Date of most recent publication	Team members	Location / institution	Papers (title, year, citekey)	What they call the approach	Key methods / fit to categorisation / AI use	Primary evidence source mode	Multi-source handling style	Maturity / status	Tooling stack
			<i>Causal Mapping</i> (2021) Powell & Ltd. (2021); <i>From Narrative Text to Causal Maps</i> (2021) Remnant (2021)						
2025	Jordan White; Pete Barbrook-Johnson	Institute for New Economic Thinking (University of Oxford); CECAN / University of Surrey	<i>Guidance on Using Large Language Models to Extract Cause-and-Effect Pairs from Texts for Systems Mapping</i> (2025) White & Barbrook-Johnson (n.d.)	Systems mapping with LLM extraction	Fit with document coding + semi-automation ; uses LLM prompts to extract cause-effect pairs then build preliminary maps for later human refinement	Documents/text corpora	Aggregated extraction over multiple texts; human validation step	Report/guidance	GPT-based extraction + external mapping tools (e.g., PRSM)
2024	Fran Ackermann; Colin	Project studies / management	<i>Overlooked and Underused?</i>	Causal mapping for projects	Fit with individual/group	Interviews, workshops,	Comparative and synthetic	Peer-reviewed articles	Decision Explorer / Group Explorer

Date of most recent publication	Team members	Location / institution	Papers (title, year, citekey)	What they call the approach	Key methods / fit to categorisation / AI use	Primary evidence source mode	Multi-source handling style	Maturity / status	Tooling stack
	Eden; James Alexander; Eunice Maytona-Sanchez	lineage; institutions not explicit in these records	(2024) Ackerman n & Maytoren a-Sanchez (2024) ; <i>Researching Complex Projects</i> (2016) Ackerman n & Alexander (2016) ; <i>Using Causal Mapping to Support Information Systems Development</i> (2011) Ackerman n & Eden (2011)	and IS development	elicitation, document-supported mapping, and idiographic systemic analysis ; strong continuity from 2011 to 2024; AI not central	documents	c use across sources		+ manual coding
2023	Rory Hooper; Nihit Goyal; Kornelis Blok; Lisa Scholten	Institution not explicit in record (policy evidence synthesis context)	<i>A Semi-Automated Approach to Policy-Relevant Evidence Synthesis</i> (2023)	Semi-automated causal mapping for policy evidence synthesis	Hybrid NLP + causal mapping + graph analytics; fit with document coding and multi-	Policy/research documents	Aggregated multi-document syntheses	Preprint	NLP pipeline + graph analytics + causal-map post-processing

Date of most recent publication	Team members	Location / institution	Papers (title, year, citekey)	What they call the approach	Key methods / fit to categorisation / AI use	Primary evidence source mode	Multi-source handling style	Maturity / status	Tooling stack
			Hooper (2023)		source synthesis; AI used directly for extraction pipeline				
2025	Philippe Giabbanelli; Tyler Gandee; Ameeta Agrawal ; Niyousha Hosseini chimeh	Applied ontology / systems mapping	<i>Benchmarking and Assessing Transformations Between Text and Causal Maps via Large Language Models</i> (2025) Giabbane lli et al. (2025)	Text-to-map and map-to-text for causal maps	Benchmarking and evaluation on datasets for LLM transformation between prose and causal maps; AI core method	Documents and causal-map corpora	Multi-dataset benchmark aggregation	Peer-reviewed article	LLMs + benchmark notebooks/metrics
2025	Melissa Valdivia Cabrera; Michael Johnstone; Joshua Hayward; Kristy Bolton; Douglas	Community health systems modelling	<i>Integration of Large-Scale Community-Developed Causal Loop Diagrams</i>	NLP-assisted CLD factor merging	Semantic similarity / NLP used to merge community-generated CLD factors	Participatory CLDs + associated text labels	Aggregated map integration across communities	Peer-reviewed articles	NLP semantic matching + network analysis /DEMA TEL

Date of most recent publication	Team members	Location / institution	Papers (title, year, citekey)	What they call the approach	Key methods / fit to categorisation / AI use	Primary evidence source mode	Multi-source handling style	Maturity / status	Tooling stack
	Creighton		ms... <i>NLP Approach</i> (2025) Valdivia Cabrera et al. (2025); foundational merge workflow in Hayward et al. (2020)		at scale; strong fit with multi-source map integration				
2024	Raquel Buzogany; Birgit Kopainsky; Paulo Goncalves	System Dynamics Review / policy narrative analysis	<i>Developing Theoretically Grounded Causal Maps to Examine and Improve Policy Narratives</i> (2024) Buzogany et al. (2024)	Theoretically grounded causal maps / CLDs	Grounded-theory coding from qualitative corpora into CLDs; fit with document coding and policy narrative synthesis	Policy and academic texts	Aggregated cross-domain causal syntheses	Peer-reviewed article	Qualitative coding + CLD modelling
2024	Mohammad S. Jalali; Ali Akhavan	System Dynamics Review	<i>Integrating AI Language Models in Qualitative</i>	AI-assisted replication of interview analysis	ChatGPT-assisted replication of interview-to-CLD	Interview transcripts	Replication against prior coded analyses	Peer-reviewed article	ChatGPT-assisted qualitative coding

Date of most recent publication	Team members	Location / institution	Papers (title, year, citekey)	What they call the approach	Key methods / fit to categorisation / AI use	Primary evidence source mode	Multi-source handling style	Maturity / status	Tooling stack
			<i>Research (2024)</i> Jalali & Akhavan (2024)		analysis; AI used as augmentative analyst				
2023	Charlotte Matthews; Will Airey; Fiona Remnant; Aurelie Charles	University of Bath / local SDG evaluation context	<i>The Dynamics of the UN Voluntary Local Review Using Causal Mapping...</i> (2023) Matthews et al. (2023)	Causal mapping for SDG VLR analysis	Cross-SDG causal mapping to identify leverage points and stakeholders in local policy; fit with document + stakeholder synthesis	VLR documents + stakeholder evidence	Within-goal and across-goal map synthesis	Applied report	Causal Map workflow + policy analysis
2022	Pete Barbrook-Johnson; Alexandra Penn; Helen Wilkins; Dione Hills (overlapping collaborators)	UK policy/evaluation systems-mapping community (institutions not always explicit in these records)	<i>Participatory Systems Mapping for Complex Energy Policy Evaluation</i> (2021) Barbrook-Johnson & Penn (2021); <i>Building a</i>	Participatory systems mapping; system-based ToC	Strong fit with group map-building, open elicitation, and translation from cyclic systems maps to evaluable ToC	Group workshops (plus some supporting docs)	Consensus-built maps with subgroup/submap analysis	Peer-reviewed + handbook chapters + practice case	Workshop facilitation + network/submap analysis

Date of most recent publication	Team members	Location / institution	Papers (title, year, citekey)	What they call the approach	Key methods / fit to categorisation / AI use	Primary evidence source mode	Multi-source handling style	Maturity / status	Tooling stack
			<i>System-Based Theory of Change Using Participatory Systems Mapping</i> (2021) Wilkinson et al. (2021); <i>Participatory Systems Mapping</i> (2022) Barbrook-Johnson & Penn (2022); <i>Running Systems Mapping Workshops</i> (2022) Barbrook-Johnson & Penn (2022); <i>Participatory Systems Mapping in Action</i> (2020) Mapping & Incentive (2020)		submaps ; AI generally not central in core 2021-2022 method papers				

Date of most recent publication	Team members	Location / institution	Papers (title, year, citekey)	What they call the approach	Key methods / fit to categorisation / AI use	Primary evidence source mode	Multi-source handling style	Maturity / status	Tooling stack
2022	Jie Yang; Soyeon Caren Han; Josiah Poon	AI-NLP causality extraction community	<i>A Survey on Extraction of Causal Relations from Natural Language Text</i> (2022) Yang et al. (2022)	Causal relation extraction from text	Survey of knowledge-based, ML, and deep-learning pipelines for extracting cause-effect links; strong document coding / text-to-causal-link relevance	Text corpora	Cross-dataset method syntheses	Peer-reviewed article	NLP extraction pipeline
2020	Luke Craven (method lineage)	Systems /evaluation research	<i>System Effects: A Hybrid Methodology for Exploring the Determinants of Food In/Security</i> (2017) Craven (2017) ; <i>Improvi</i>	System Effects (hybrid SSM + FCM + graph analysis)	Fit with participant-derived maps + aggregate structural analysis across cases; no direct AI core component	Interviews/workshops + syntheses	Aggregated comparative map structures	Peer-reviewed + applied report	FCM + graph-theoretic analysis

Date of most recent publication	Team members	Location / institution	Papers (title, year, citekey)	What they call the approach	Key methods / fit to categorisation / AI use	Primary evidence source mode	Multi-source handling style	Maturity / status	Tooling stack
			<i>ng the Health, Wellbeing...</i> (2020) Craven (2020)						
2020	Steven E. Wallis (plus prior collaborators in IPA lineage)	Institution not explicit in record	<i>Integrative Propositional Analysis for Developing Capacity...</i> (2020) Wallis (2020)	Integrative Propositional Analysis (IPA)	Causal propositions mapped and scored structurally (breadth/systemicity); fits analysis-heavy map evaluation rather than pure elicitation; AI not central	Strategic documents/propositions	Integrated conceptual synthesis	Peer-reviewed article	Manual proposition extraction + structural metrics
2020	Sasha Strelnikoff; Aruna Jammalamadaka; Dana Warmusley	Institution not explicit in record	<i>Causal Maps for Multi-Documents Summarization</i> (2020) Strelnikoff et	Causal maps for multi-document summarization	Fully unsupervised NLP pipeline for cause-effect extraction and clustering; fits	Multi-document corpora	Aggregated large-scale document synthesis	Peer-reviewed conference paper	DeepCx + embeddings + mixture model + graph pruning

Date of most recent publication	Team members	Location / institution	Papers (title, year, citekey)	What they call the approach	Key methods / fit to categorisation / AI use	Primary evidence source mode	Multi-source handling style	Maturity / status	Tooling stack
			al. (2020)		document coding at scale with explicit AI/NLP core				
2018	Igor Pyrko; Viktor Dorfler	Management/organization research context	<i>Using Causal Mapping in the Analysis of Semi-structured Interviews</i> (2018) Pyrko & Dorfler (2018)	Eden/Ackerman-style causal mapping for interview analysis	Strong fit with interview coding, map merging across respondents , and feedback-chain analysis ; AI not central	Semi-structured interviews	Merged intersubjective maps from individual interview maps	Conference proceedings paper	Manual coding + map-merging analysis
2018	Ricardo Wilson-Grau; Heather Britt	Outcome Harvesting practice community	<i>Outcome Harvesting Principles in Practice</i> (2018) Wilson-Grau (2018); <i>Outcome Harvesting</i> (2012) Wilson-	Outcome Harvesting (adjacent causal-claim approach)	Adjacent method for harvesting causal contribution claims from narrative evidence ; fits multi-source	Narrative outcomes/interviews/documents	Multi-source claim harvesting	Guidance/practice documents	Manual harvesting/coding

Date of most recent publication	Team members	Location / institution	Papers (title, year, citekey)	What they call the approach	Key methods / fit to categorisation / AI use	Primary evidence source mode	Multi-source handling style	Maturity / status	Tooling stack
			Grau & Britt (2012)		evidence assembly; typically no AI core				
2017	Gerard Hodgkinson; Kristian Sund; Robert Galavan	Managerial and organizational cognition community	Chapter 1: <i>Exploring Methods in Managerial and Organizational Cognition</i> (2017) Hodgkins on et al. (2017)	Causal/cognitive mapping methods in MOC	Methodological syntheses of causal mapping in management cognition; mainly interview/document elicitation + map analysis tradition; AI not central	Interviews/documents (methodological review)	Comparative methodological syntheses	Book chapter	Method framework syntheses
2016	Mauri Laukkanen; Mingde Wang; Päivi Eriksson	Comparative causal mapping lineage (management/organization)	<i>Comparative Causal Mapping: The CMAP3 Method</i> (2016) Laukkane n & Wang (2016); <i>New</i>	Comparative Causal Mapping (CCM), CMAP3	Core comparative/aggregate map methodology , including standardized	Interviews/documents (structured to low-structured variants)	Comparative aggregation across multiple respondents/cases	Book + peer-reviewed articles	CMAP3 + manual standardization

Date of most recent publication	Team members	Location / institution	Papers (title, year, citekey)	What they call the approach	Key methods / fit to categorisation / AI use	Primary evidence source mode	Multi-source handling style	Maturity / status	Tooling stack
			<i>Designs and Software for Cognitive Causal Mapping</i> (2013) Laukkane & Eriksson (2013); <i>Comparative Causal Mapping and CMAP3 Software in Qualitative Studies</i> (2012) Laukkane (2012)		concept pools and software - supported cross-case comparison; AI not central				
2014	Michal Sedlacko; Andre Martinuzzi; Inge Ropke; Nuno Videira; Paula Antunes	Sustainability / ecological economics context	<i>Participatory Systems Mapping for Sustainable Consumption</i> (2014) Sedlacko et al. (2014)	Participatory systems mapping	Early strong participatory systems-mapping method paper; fits group map-building and systemic insight generation; AI	Participatory workshops	Group-built causal map syntheses	Peer-reviewed article	Workshop methods + map analysis

Date of most recent publication	Team members	Location / institution	Papers (title, year, citekey)	What they call the approach	Key methods / fit to categorisation / AI use	Primary evidence source mode	Multi-source handling style	Maturity / status	Tooling stack
					not central				

Notes on team boundaries

- Teams are grouped by overlapping authors and clear method lineage; single-paper rows are kept where overlap is weak or absent.
- Duplicates were collapsed where records represent the same output (for example (n.d.) vs (n.d.)).
- Obvious noisy keyword matches (items tagged causal mapping but not materially about causal-mapping methods) were excluded.

From your supplied list: explicitly not in this table (and why)

- **Pre-2011 foundational classics** (kept out only due to 15-year scope): e.g., Axelrod 1976, Eden 1988/1992, Eden/Ackermann/Cropper 1992, Narayanan/Armstrong 2004, Clarkson/Hodgkinson 2005.
- **Outside causal-mapping core despite causal relevance:** e.g., Pearl 2000, Forrester 1971, Tolman 1948, Wright 1921 (important background, but not causal-mapping method teams).
- **General AI/qualitative-method papers with weak direct mapping focus:** retained only when explicitly tied to causal-map/CLD production or transformation.

Related

- [chapter intro](#)

References

Ackermann, & Eden (2011). *Using Causal Mapping to Support Information Systems Development*.

Ackermann, & Alexander (2016). *Researching Complex Projects: Using Causal Mapping to Take a Systems Perspective*. <https://linkinghub.elsevier.com/retrieve/pii/S0263786316300072>.

Ackermann, & Maytorena-Sanchez (2024). *Overlooked and Underused? The Benefits and Challenges of Using Causal Mapping for Project Studies*. <https://doi.org/10.1016/j.plas.2024.100161>.

- Avard, Mukuru, & Liesner (2022). *Measuring the Women's Economic Empowerment Generated by Impact Investing; Testing the QuIP Method on an Investment in Uganda's Cotton Sector*. Taylor & Francis.
- Barbrook-Johnson, & Penn (2021). *Participatory Systems Mapping for Complex Energy Policy Evaluation*. <http://dx.doi.org/10.1177/1356389020976153>.
- Barbrook-Johnson, & Penn (2022). *Running Systems Mapping Workshops*. In *Systems Mapping: How to Build and Use Causal Models of Systems*. https://doi.org/10.1007/978-3-031-01919-7_10.
- Barbrook-Johnson, & Penn (2022). *Participatory Systems Mapping*. In *Systems Mapping: How to Build and Use Causal Models of Systems*. https://doi.org/10.1007/978-3-031-01919-7_5.
- Britt, Powell, & Cabral (2025). *Strengthening Outcome Harvesting with AI-assisted Causal Mapping*. https://5a867cea-2d96-4383-acf1-7bc3d406cdeb.usrfiles.com/ugd/5a867c_ad000813c80747baa85c7bd5ffafo442.pdf.
- Buzogany, Kopainsky, & Gonçalves (2024). *Developing Theoretically Grounded Causal Maps to Examine and Improve Policy Narratives about Global Challenges*. <https://doi.org/10.1002/sdr.1788>.
- Craven (2017). *System Effects: A Hybrid Methodology for Exploring the Determinants of Food In/Security*. <https://www.tandfonline.com/doi/full/10.1080/24694452.2017.1309965>.
- Craven (2020). *Improving the Health, Wellbeing , and Chronic Disease Management of the Arabic Speaking Community Data - Through the Culture Well Project , Asthma Australia*.
- Giabbanelli, Gandee, Agrawal, & Hosseinichimeh (2025). *Benchmarking and Assessing Transformations Between Text and Causal Maps via Large Language Models*. SAGE Publications. <https://doi.org/10.1177/15705838241304102>.
- Hammond, Lea, & Hammond (2026). *Where to Start? Participatory Systems Mapping for Place-Based Service Integration in the City of Casey*. <https://doi.org/10.20944/preprints202601.2371.v1>.
- Hayward, Morton, Johnstone, Creighton, & Allender (2020). *Tools and Analytic Techniques to Synthesise Community Knowledge in CBPR Using Computer-Mediated Participatory System Modelling*. <https://www.nature.com/articles/s41746-020-0230-x>.
- Hodgkinson, Sund, & Galavan (2017). *Chapter 1: Exploring Methods in Managerial and Organizational Cognition: Advances, Controversies, and Contributions*. In *New Horizons in Managerial and Organizational Cognition*. <https://doi.org/10.1108/S2397-52102017002>.
- Hooper (2023). *A Semi-Automated Approach to Policy-Relevant Evidence Synthesis: Combining Natural Language Processing, Causal Mapping, and Graph Analytics for Public Policy*. <https://doi.org/10.21203/rs.3.rs-3285731/v1>.
- Jalali, & Akhavan (2024). *Integrating AI Language Models in Qualitative Research: Replicating Interview Data Analysis with ChatGPT*. <https://doi.org/10.1002/sdr.1772>.
- Laukkanen (2012). *Comparative Causal Mapping and CMAP3 Software in Qualitative Studies*. <https://doi.org/10.17169/fqs-13.2.1846>.

- Laukkanen, & Eriksson (2013). *New Designs and Software for Cognitive Causal Mapping*.
<https://www.emerald.com/insight/content/doi/10.1108/QROM-08-2011-1003/full/html>.
- Laukkanen, & Wang (2016). *Comparative Causal Mapping: The CMAP3 Method*. Routledge.
- Mapping, & Incentive (2020). *Participatory Systems Mapping in Action —Supporting the Evaluation of the Renewable Heat Incentive*.
- Matthews, Airey, Remnant, & Charles (2023). *The Dynamics of the UN Voluntary Local Review Using Causal Mapping within and across the Sustainable Development Goals: A Case Study of Bath and North East Somerset*.
- Powell, & Ltd. (2021). *Guide to Causal Mapping*. <https://info.causalmap.app/>.
- Powell, Remnant, Copestake, Mishan, Arvard, & Goddard (2022). *Chapter 1 Overview | Guide to Causal Mapping*. <https://causalmap.app>.
- Powell, Larquemin, Copestake, Remnant, & Avarð (2023). *Does Our Theory Match Your Theory? Theories of Change and Causal Maps in Ghana*. In *Strategic Thinking, Design and the Theory of Change. A Framework for Designing Impactful and Transformational Social Interventions*.
- Powell, Cabral, & Mishan (2024). *An M\&E Time Machine: Using AI to Measure Changes in a System across a Time Period on Features Which Only Emerge during It..*
<https://doi.org/10.31124/advance.171535113.38566027/v1>.
- Powell, Copestake, & Remnant (2024). *Causal Mapping for Evaluators*.
<https://doi.org/10.1177/13563890231196601>.
- Pyrko, & Dorfler (2018). *Using Causal Mapping in the Analysis of Semi-structured Interviews*.
<https://doi.org/10.5465/AMBPP.2018.14348abstract>.
- Remnant (2021). *From Narrative Text to Causal Maps- QuIP Analysis and Visualisation 24 Oct 2021*.
- Remnant, Copestake, Powell, & Channon (2025). *Qualitative Causal Mapping in Evaluations*. In *Handbook of Health Services Evaluation: Theories, Methods and Innovative Practices*.
https://doi.org/10.1007/978-3-031-87869-5_12.
- Sedlacko, Martinuzzi, Røpke, Videira, & Antunes (2014). *Participatory Systems Mapping for Sustainable Consumption: Discussion of a Method Promoting Systemic Insights*. Elsevier.
- Strelnikoff, Jammalamadaka, & Warmesley (2020). *Causal Maps for Multi-Document Summarization*.
<https://doi.org/10.1109/BigData50022.2020.9377731>.
- Valdivia Cabrera, Johnstone, Hayward, Bolton, & Creighton (2025). *Integration of Large-Scale Community-Developed Causal Loop Diagrams: A Natural Language Processing Approach to Merging Factors Based on Semantic Similarity*. <https://doi.org/10.1186/s12889-025-22142-3>.
- Wallis (2020). *Integrative Propositional Analysis for Developing Capacity in an Academic Research Institution by Improving Strategic Planning*. <https://doi.org/10.1002/sres.2599>.
- White, & Barbrook-Johnson (n.d.). *Guidance on Using Large Language Models to Extract Cause-and-Effect Pairs from Texts for Systems Mapping* Jordan White¹ and Pete Barbrook-Johnson^{1,2} ¹

Institute for New Economic Thinking, University of Oxford 2 Centre for the Evaluation of Complexity Across the Nexus, University of Surrey.

Wilkinson, Hills, Penn, & Barbrook-Johnson (2021). *Building a System-Based Theory of Change Using Participatory Systems Mapping*. SAGE Publications Ltd.

<https://doi.org/10.1177/1356389020980493>.

Wilson-Grau, & Britt (2012). *Outcome Harvesting*.

Wilson-Grau (2018). *Outcome Harvesting: Principles, Steps, and Evaluation Applications*. IAP.

Yang, Han, & Poon (2022). *A Survey on Extraction of Causal Relations from Natural Language Text*.

<https://doi.org/10.1007/s10115-022-01665-w>.